

Data Analyst Assessment | VirtuBox

Marketplace Logistics & Customer Retention Diagnosis

Focus: Investigating delivery SLA compliance, review collapse, and cohort churn across 100k+ orders.

Candidate: **Tapish Kumar**

Data Scope & Analytical Pipeline

- Dataset: Olist Brazilian E-Commerce Public Dataset (99,441 orders, 112,650 items across 9 relational tables).
- Data Pipeline: Cleaned using Python/Pandas; isolated valid delivered orders and filtered corrupted records.
- Key Derivations: Extracted transit lead times, SLA variance (delay days), and re-mapped customer cohorts using `customer_unique_id` to evaluate true repeat retention.

Executive Platform Health Summary

- Total Marketplace GMV: \$13.59M across 73 product categories.
- On-Time Delivery Rate: 91.9% platform baseline.
- Repeat Buyer Deficit: 3.12% repeat rate (96.88% of users buy only once, exposing reliance on paid acquisition).
- Customer Satisfaction: 4.08 / 5.0 baseline average review score.

Deep Dive: Fulfillment SLA vs. CSAT Collapse

- Delivery Delay Impact: On-time orders average 4.27/5.0 stars; delayed orders drop 60.6% to 1.68/5.0 stars.
- Surprising Finding: Lightweight electronics experience a 14.2% delay rate compared to 7.8% for bulky furniture.
- Algorithmic Flaw: Flaws in delivery promise calculations create unrealistic 3-5 day windows on small parcels, driving customer disappointment rather than courier failures.

Strategic Recommendations

1. Dynamic SLA Buffer Calibration: Retrain delivery estimate engines on 85th-percentile regional transit history rather than flat averages.
2. Automated 21-Day CRM Cross-Sell: Trigger tailored category re-engagement campaigns within 3 weeks of confirmed fulfillment.
3. Regional 3PL Cross-Docking: Partner with local logistics providers in Northeast hubs to reduce 21-28 day delivery times.

Expected Business Impact & Success Metrics

- Retention Lift: Double platform repeat buyer rate from 3.12% to 6.0% within 6 months.
- CSAT Recovery: Reduce SLA breaches by 45%, raising overall platform rating to 4.35+.
- Core Tracked Metrics: SLA breach rate (<4%), transit duration across interstate routes, and 60-day cohort retention.

Analytical Limitations & Next Steps

- Analytical Limitations: Lacks advertising spend (CAC) and merchant cost-of-goods-sold (COGS), restricting unit economics to GMV.
- Next Steps: Ingest pre-purchase funnel telemetry (cart drop-offs) and carrier shipping invoice audits to model net EBITDA per order.